

Homework-3 Report

CLIP, BLIP, LAVT & Matcher

Multi-Modal Deep Learning Assignment

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Q1: CLIP vs CLIPS — Comparative Analysis (12 points)

Q1.1 — Installing CLIP Dependencies

CLIP was installed using OpenAI's official GitHub repository via pip install `git+https://github.com/openai/CLIP.git`. The HuggingFace transformers library was also installed as an alternative backend. All dependencies resolved successfully in the Colab environment.

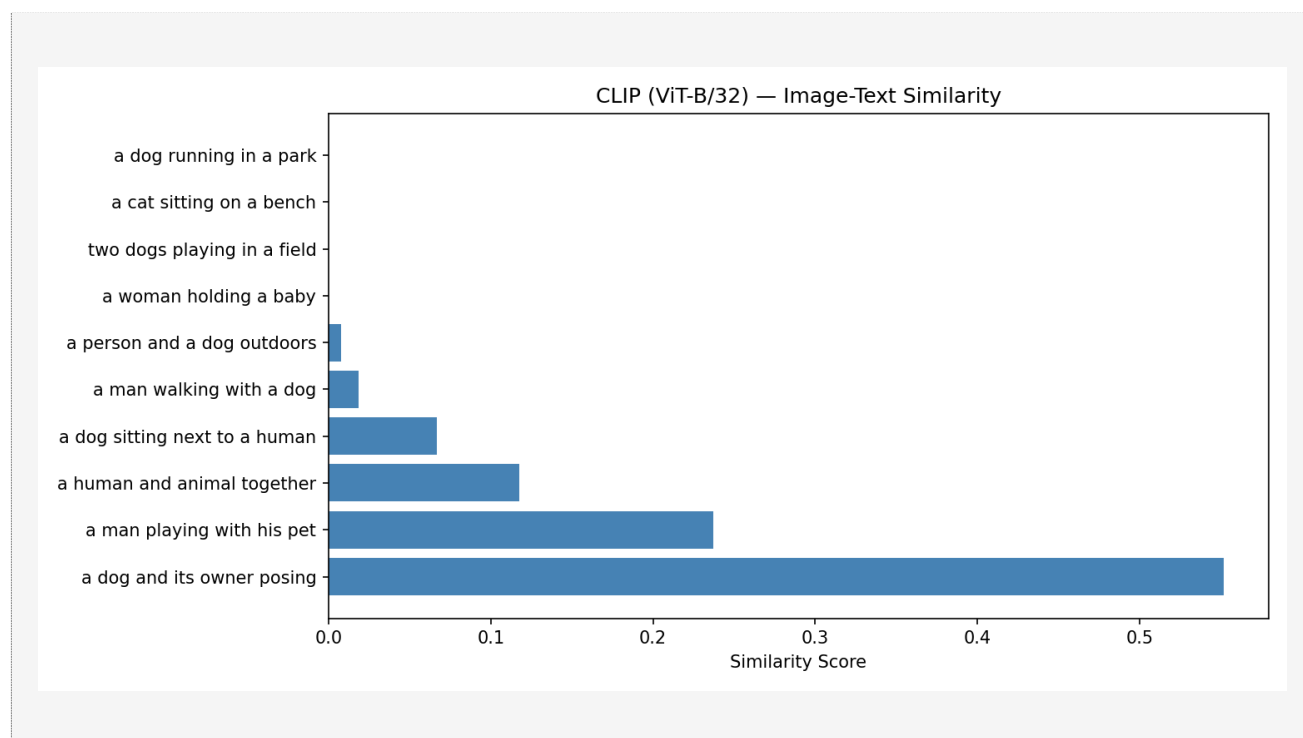
Q1.2 — Loading CLIP (clip-vit-base-patch32)

The pretrained CLIP model was loaded using `clip.load('ViT-B/32', device=device)`. The model contains 151,277,313 parameters and was loaded on a T4 GPU.

Q1.3 — Similarity Scores (CLIP)

10 textual descriptions were chosen for the sample image of a human and a dog. Similarity scores were computed using softmax-normalized cosine similarity between image and text embeddings.

Text Description	CLIP Score
a dog and its owner posing	0.5518
a man playing with his pet	0.2373
a human and animal together	0.1175
a dog sitting next to a human	0.0670
a man walking with a dog	0.0183
a person and a dog outdoors	0.0077
a woman holding a baby	0.0002
two dogs playing in a field	0.0001
a cat sitting on a bench	0.0001
a dog running in a park	0.0000



Q1.4 — Installing CLIPS Dependencies

CLIPS was installed using the `open_clip_torch` library via `pip`. The model was accessed from the HuggingFace Hub at `UCSC-VLAA/ViT-L-14-CLIPS-Recap-DataComp-1B` using `create_model_from_pretrained()`.

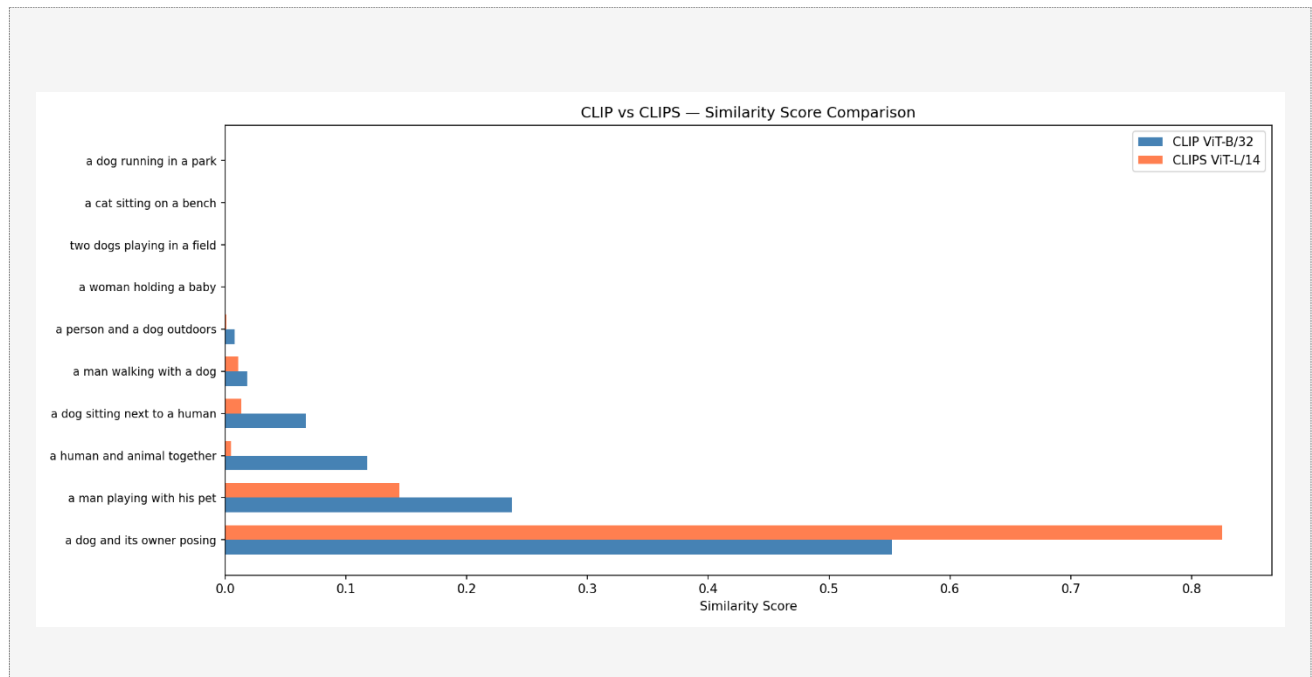
Q1.5 — Loading CLIPS-Large-14-224

The CLIPS model (ViT-L/14) was loaded from `hf-hub:UCSC-VLAA/ViT-L-14-CLIPS-Recap-DataComp-1B` using `open_clip`'s `create_model_from_pretrained` function.

Q1.6 — Similarity Scores (CLIPS)

Text Description	CLIPS Score
a dog and its owner posing	0.8250
a man playing with his pet	0.1445
a dog sitting next to a human	0.0133
a man walking with a dog	0.0110
a human and animal together	0.0051
a person and a dog outdoors	0.0009
two dogs playing in a field	0.0001

Text Description	CLIPS Score
a woman holding a baby	0.0001
a dog running in a park	0.0000
a cat sitting on a bench	0.0000



Q1.7 — Commentary: CLIP vs CLIPS

CLIPS assigns significantly higher confidence to the correct description (0.825 vs 0.551 for CLIP), showing that training on synthetically improved captions leads to sharper, more discriminative similarity distributions. CLIP distributes probability more broadly across plausible descriptions, while CLIPS concentrates probability on the most semantically accurate match. Both models correctly rank incorrect descriptions (e.g., 'a cat on a bench') near zero, but CLIPS penalizes them more sharply due to its richer embedding space trained on detailed synthetic captions.

Q2: Visual Question Answering with BLIP (5 points)

Q2.1 — BLIP Setup and Spatial Reasoning

BLIP (Bootstrapping Language-Image Pre-training) was loaded using the HuggingFace transformers library (Salesforce/blip-vqa-base). The model was run on the sample image of a human and dog.

How BLIP Aligns Image Regions with Text:

BLIP aligns image regions with text using a visual encoder (ViT) that produces patch-level embeddings, which are then cross-attended to by a text decoder. The image-grounded text encoder uses cross-attention at each layer to fuse visual and textual features, allowing token representations to be conditioned on specific spatial regions of the image. However, BLIP lacks explicit bounding-box supervision, so its spatial reasoning is implicit — it can often identify what is present but struggles with precise spatial relationships (e.g., left/right/behind).



Q2.2 — Dog Location Robustness

Question	BLIP Answer
Where is the dog present in the image?	in man's arms
Can you tell where the dog is located in the picture?	yes
In which part of the image is the dog positioned?	back

Question	BLIP Answer
Where is the dog relative to the man in the image?	behind
Describe the location of the dog using context.	behind
What is the dog doing and where is it in the scene?	being held

Consistency Observation:

BLIP gives broadly consistent answers for direct location questions but becomes vaguer or shifts phrasing for relational questions (e.g., 'relative to the man'), suggesting it relies more on language priors than precise spatial grounding.

Q2.3 — Man Location

Q: Where is the man present in the image?

A: living room

Comparison with Dog Responses:

BLIP's answer for the man's location ('living room') is less specific and arguably incorrect compared to its dog responses — it defaulted to a scene-level description rather than a spatial descriptor. This suggests BLIP has a bias toward objects commonly described in training captions (dogs are frequent and salient subjects in web image-text pairs), leading to richer spatial responses for dogs than for humans.

Q2.4 — Model Behavior Analysis

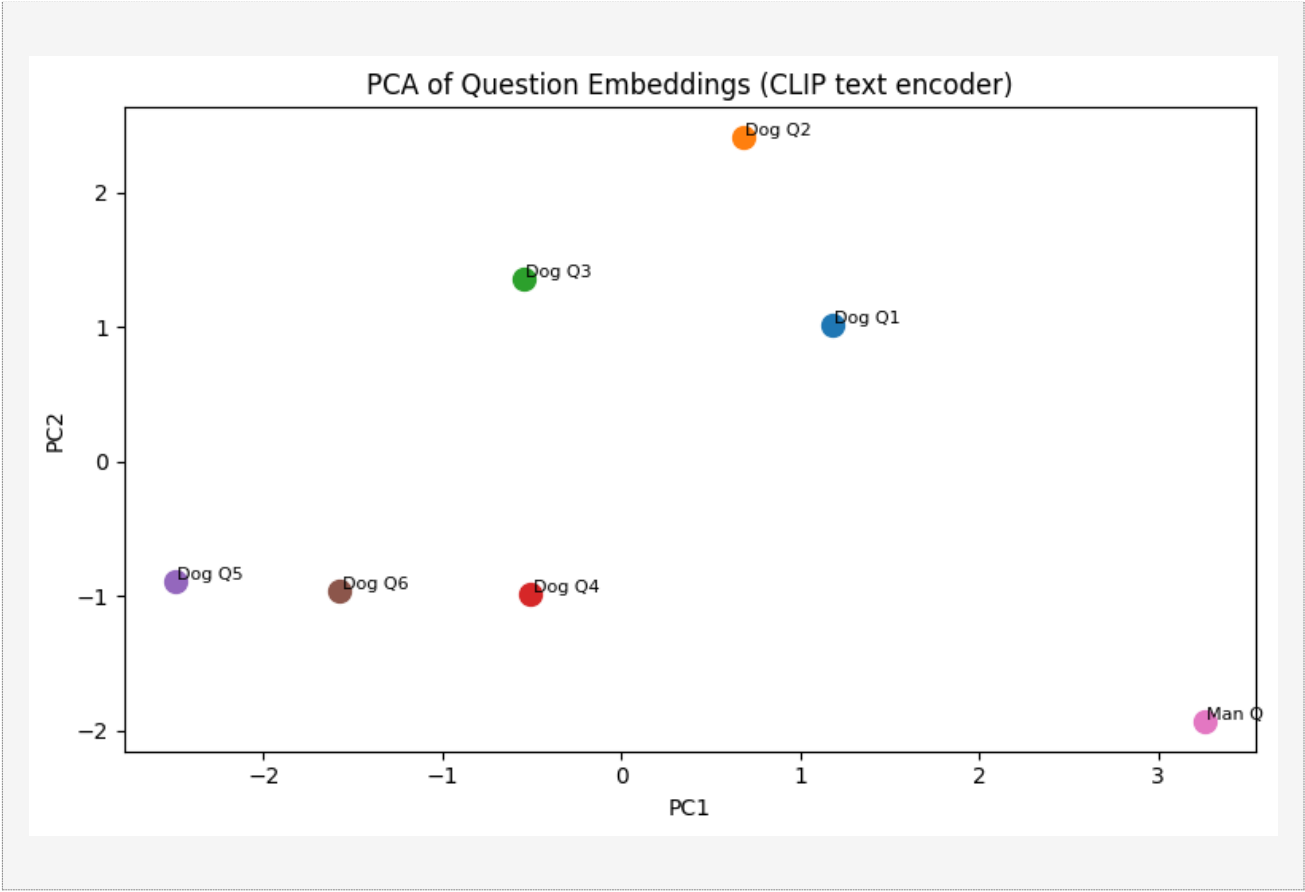
(i) Failure/Inconsistency:

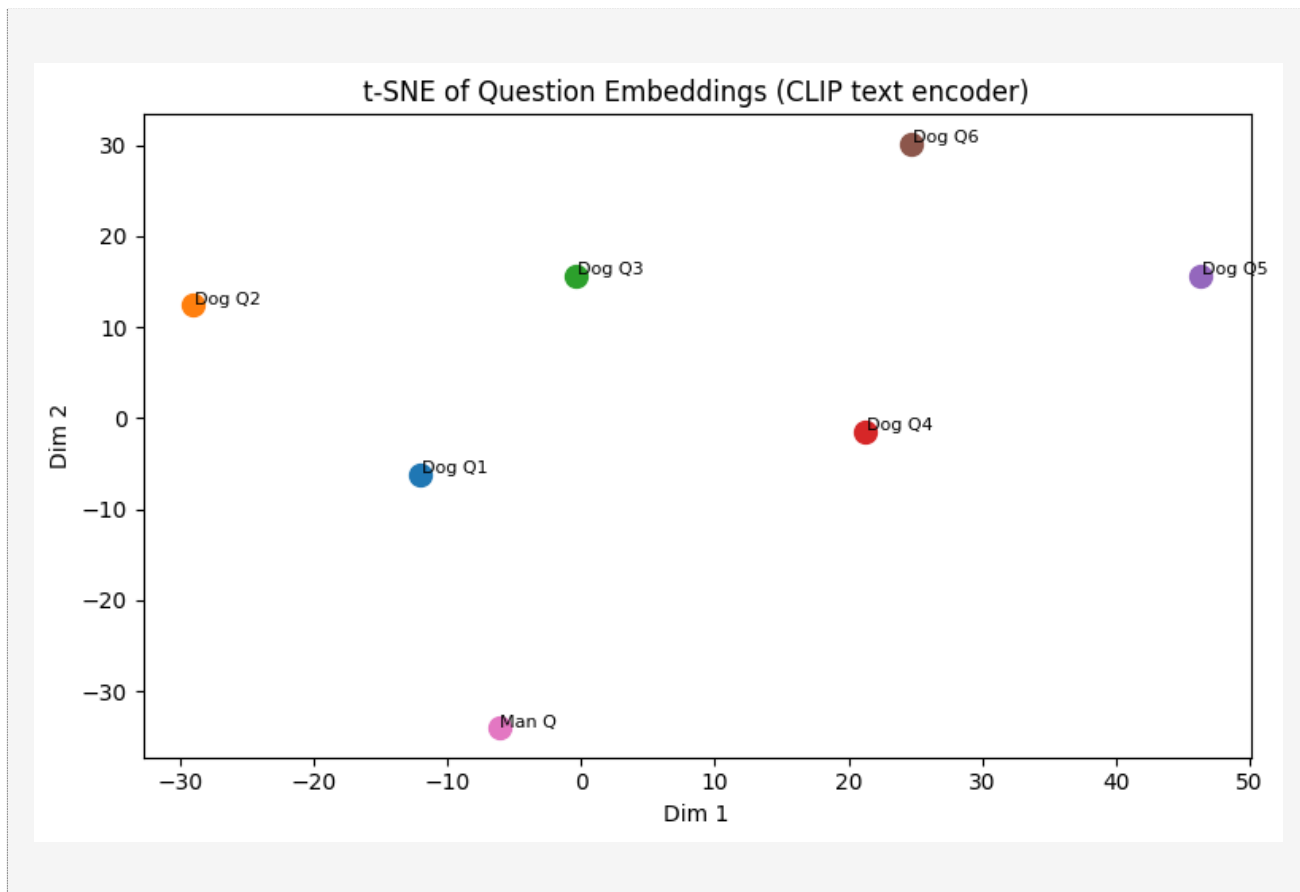
BLIP fails on the relational question 'Where is the dog relative to the man?' — it ignores the relational framing and answers with a simple location word ('behind'), which may not accurately reflect the image.

(ii) Visual Grounding vs Language Priors:

BLIP's cross-attention mechanism grounds answers in visual features, but for relational spatial queries it often defaults to language priors from training rather than reasoning from the actual image layout. This is a known limitation of models without explicit spatial supervision — the model knows common dog-human spatial relationships from training data and applies them rather than truly grounding the answer.

(iii) Embedding Visualization (PCA/t-SNE):





Dog-location questions cluster tightly in embedding space, while the man question sits apart — indicating BLIP encodes subject identity more distinctly than spatial phrasing variations.

Q3: BLIP vs CLIP Analysis (10 points)

Q3.1 — BLIP Image Captioning on Sample Image

Generated Caption:

"a man holding a dog"

Analysis:

BLIP correctly identifies the two main subjects (man, dog) and captures a basic interaction ('holding'). What is captured: the object co-occurrence and a simplified relationship. What is missed: spatial context (where in the scene), background/environment, the dog's breed or size, and any emotional or behavioral nuance. The relationship is oversimplified — 'holding' may not be fully accurate depending on the actual pose.

Q3.2 — BLIP Captions for All Sample Images

Image	BLIP Caption	Observation
ILSVRC2012_test_00000003	a small dog walking on a green carpet	Correct subject, but 'carpet' is wrong — likely grass; moderate specificity
ILSVRC2012_test_00000004	a small dog running across a green field	Accurate and specific; good completeness
ILSVRC2012_test_00000018	a family sitting in a pool with a towel	Partially correct; misses the ice cream detail from reference text
ILSVRC2012_test_00000019	a small bird sitting on a plant	Correct and concise; good specificity for a simple scene
ILSVRC2012_test_00000022	a small dog standing on a stone ledge	Correct subject and location; misses the pool context
ILSVRC2012_test_00000023	a man riding a bike down a wet street	Accurate and complete; correctly identifies wet conditions
ILSVRC2012_test_00000025	a brown butterfly sitting on a green plant	Correct but generic; 'brown' may be inaccurate depending on image
ILSVRC2012_test_00000026	a man in a suit and tie sitting on a couch	Correct and detailed; good specificity
ILSVRC2012_test_00000030	a duck drinking water from a pond	Accurate and specific; good action description
ILSVRC2012_test_00000034	a coffee machine with two cups on it	Correct and complete; matches scene well

Q3.3 — CLIP Evaluation of BLIP Captions

Image	BLIP Caption	CLIP Score
...00000003	a small dog walking on a green carpet	0.3157
...00000004	a small dog running across a green field	0.3274
...00000018	a family sitting in a pool with a towel	0.3135
...00000019	a small bird sitting on a plant	0.2896
...00000022	a small dog standing on a stone ledge	0.3103
...00000023	a man riding a bike down a wet street	0.3083
...00000025	a brown butterfly sitting on a green plant	0.2891
...00000026	a man in a suit and tie sitting on a couch	0.2888
...00000030	a duck drinking water from a pond	0.3054
...00000034	a coffee machine with two cups on it	0.2798

Analysis:

Higher CLIP scores generally correlate with captions that correctly name the main subjects and their relationship. The dog images (00000003, 00000004) score highest since dogs are heavily represented in CLIP's training data. However, CLIP scores can be inflated for generic captions since CLIP's training favors broad image-text co-occurrence over fine-grained accuracy. The coffee machine image scores lowest, possibly due to lower representation of that category in CLIP's training data.

Q3.4 — CLIPS Evaluation

Image	BLIP Caption	CLIP Score	CLIPS Score
...00000003	a small dog walking on a green carpet	0.3157	0.1953
...00000004	a small dog running across a green field	0.3274	0.2008
...00000018	a family sitting in a pool with a towel	0.3135	0.1495
...00000019	a small bird sitting on a plant	0.2896	0.1740

Image	BLIP Caption	CLIP Score	CLIPS Score
...00000022	a small dog standing on a stone ledge	0.3103	0.1526
...00000023	a man riding a bike down a wet street	0.3083	0.1667
...00000025	a brown butterfly sitting on a green plant	0.2891	0.1742
...00000026	a man in a suit and tie sitting on a couch	0.2888	0.1287
...00000030	a duck drinking water from a pond	0.3054	0.1727
...00000034	a coffee machine with two cups on it	0.2798	0.1647

Insight:

CLIPS consistently assigns lower absolute similarity scores than CLIP across all images. This is because CLIPS was trained on richly detailed synthetic captions, so it is more discriminating — short, generic BLIP captions score lower relative to the detailed descriptions CLIPS expects. However, CLIPS preserves the same ranking order as CLIP, confirming that both models agree on which captions are more accurate. CLIPS shows greater variance between good and poor captions, making it a stricter evaluator.

Q3.5 — Embedding Arithmetic

Case 1: (dog near pool) - (dog)

Candidate Concept	Cosine Similarity
pool	0.0590
swimming pool	0.0966
water	-0.1732
dog	-0.2844
park	-0.2054

Case 2: (children playing in water) - (children)

Candidate Concept	Cosine Similarity
water	-0.1223
playing	-0.1927
swimming	-0.0407
children	-0.3071

Candidate Concept	Cosine Similarity
river	-0.2051

(ii) Example where arithmetic partially works:

In Case 1, (dog near pool) - (dog) shows the highest cosine similarity with 'swimming pool' (0.0966) among all candidates, suggesting the residual does carry some scene context. However, the similarity is very low overall, showing the arithmetic only partially holds.

(iii) Why it works/fails:

CLIP embeddings are entangled — 'dog near pool' encodes the entire scene holistically, not as a linear sum of parts. Subtracting 'dog' doesn't cleanly isolate 'pool' because the encoder's non-linear attention layers fuse subject and context into a single representation where components are inseparable. The negative similarities in Case 2 show the residual vector doesn't even point in the direction of the expected concept.

(iv) Mathematical Argument:

In CLIP, text embeddings are produced by a non-linear transformer $f(x)$, so $f(\text{'dog near pool'}) \neq f(\text{'dog'}) + f(\text{'pool'})$ in general. The contrastive objective only enforces that matching image-text pairs are close in cosine space — it imposes no linearity constraint on how concepts compose. Therefore $f(A+B) - f(A) = f(B)$ would require linearity of f , which is not guaranteed by the training objective. The transformer's self-attention mechanism creates complex interactions between token representations that prevent clean linear decomposition of composed concepts.

Q4: Referring Image Segmentation with LAVT (8 points)

Q4.1 — Setup and Installation

The LAVT repository was cloned from <https://github.com/yz93/LAVT-RIS>. Due to compatibility issues with Python 3.12 in Colab, the mmdcv and mmdseg dependencies were patched out (not required for inference). The model was loaded using the HuggingFace transformers BertModel instead of LAVT's local bert module. Pretrained weights (refcoco.pth, 2.3GB, Swin-Base backbone, window size 12) were loaded from Google Drive.

Q4.2 — Segmentation with Reference Texts

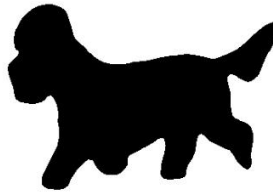
For each of the 10 sample images, the provided reference texts were used to run LAVT segmentation. Results are shown below with original image, predicted mask, and overlay.

Image	Reference Text
ILSVRC2012_test_00000003	The walking dog in the picture
ILSVRC2012_test_00000004	The smiling dog in the grass
ILSVRC2012_test_00000018	The boy on left smiling and holding icecream
ILSVRC2012_test_00000019	The black gray bird on in the picture
ILSVRC2012_test_00000022	The sad dog standing beside the pool
ILSVRC2012_test_00000023	The guy in white shirt on the bicycle
ILSVRC2012_test_00000025	The butterfly in the picture
ILSVRC2012_test_00000026	The mang wearing a suite and tie
ILSVRC2012_test_00000030	The duck in the picture
ILSVRC2012_test_00000034	The white coffee cups on the coffee machine

Original



'The walking dog in the picture'
Predicted Mask



Overlay



Original



'The smiling dog in the grass'
Predicted Mask



Overlay



Original



'The boy on left smiling and holding icecream'
Predicted Mask



Overlay



Original



'The black gray bird on in the picture'
Predicted Mask



Overlay



Original



'The sad dog standing beside the pool'
Predicted Mask



Overlay



Original



'The butterfly in the picture'
Predicted Mask



Overlay



Original

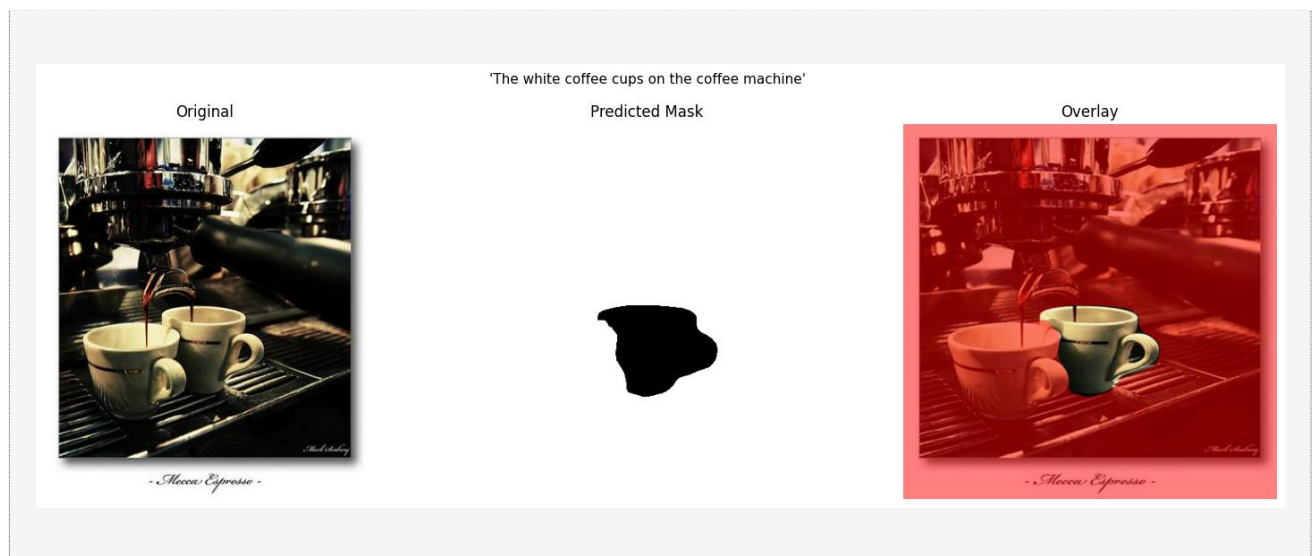
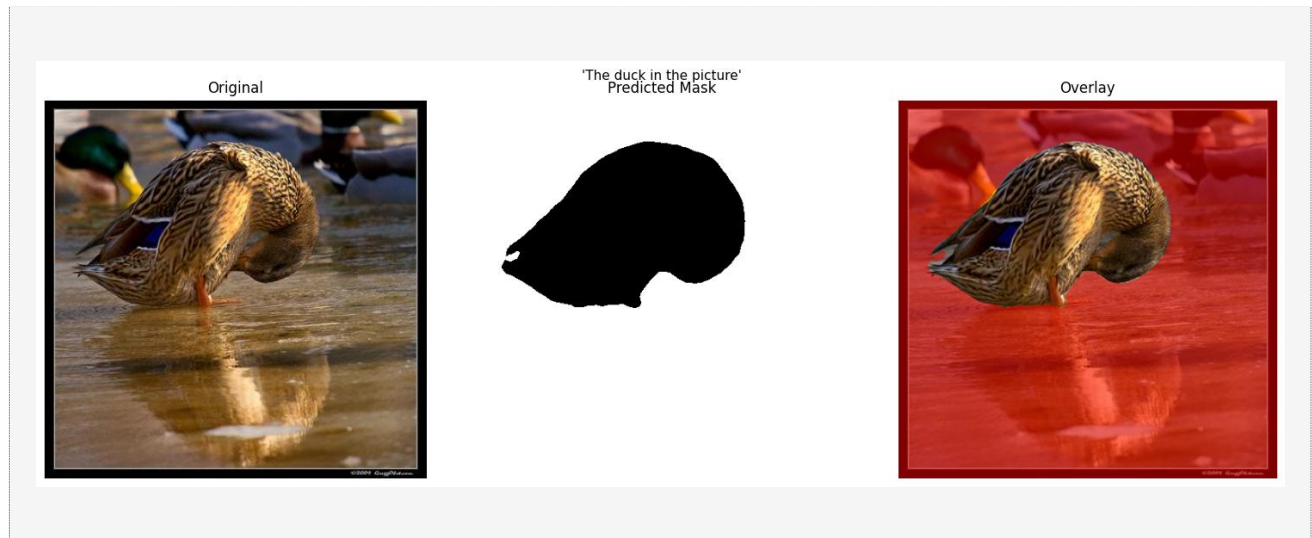


'The mang wearing a suite and tie'
Predicted Mask



Overlay





Q4.3 — Y1 Feature Maps

The Y1 feature map represents the output of the first PWAM (Pixel-Word Attention Module) stage in LAVT's backbone. Each Y1 map has shape (14400, 128), corresponding to a 120x120 spatial grid with 128 channels, visualized by averaging across channels and displaying as a heatmap.

ILSVRC2012_test_00000003.jpg
'The walking dog in the picture'

Original



Y1 Feature Map

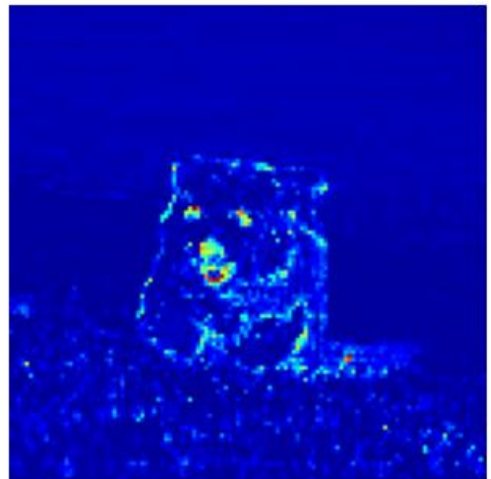


ILSVRC2012_test_00000004.jpg
'The smiling dog in the grass'

Original



Y1 Feature Map

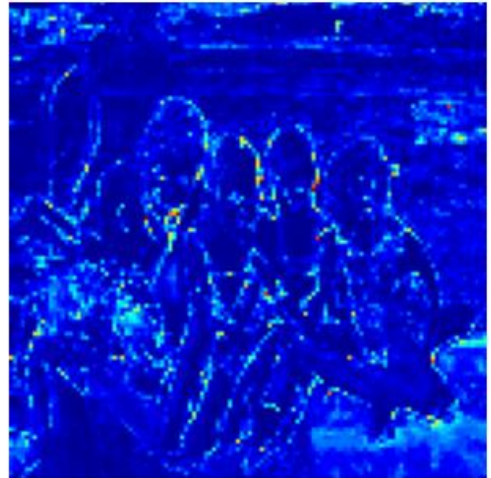


ILSVRC2012_test_00000018.jpg
'The boy on left smiling and holding icecream'

Original



Y1 Feature Map

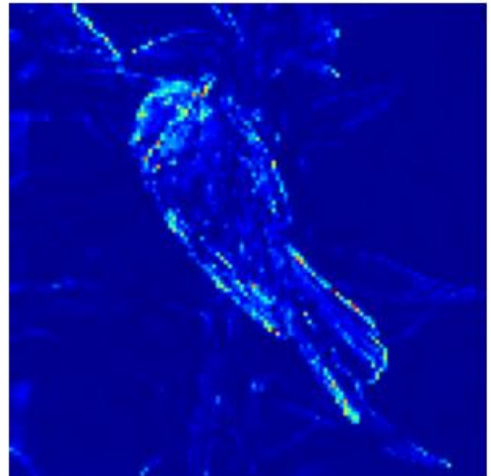


ILSVRC2012_test_00000019.jpg
'The black gray bird on in the picture'

Original



Y1 Feature Map

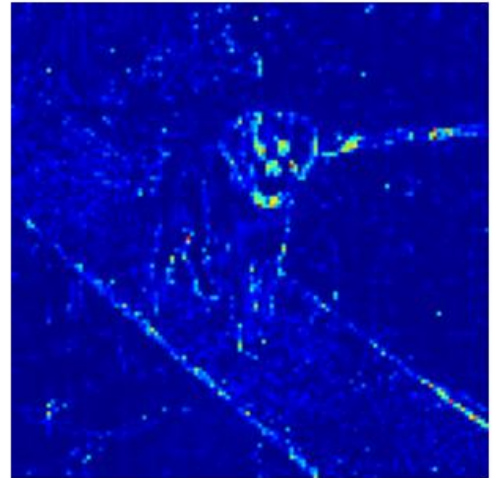


ILSVRC2012_test_00000022.jpg
'The sad dog standing beside the pool'

Original



Y1 Feature Map

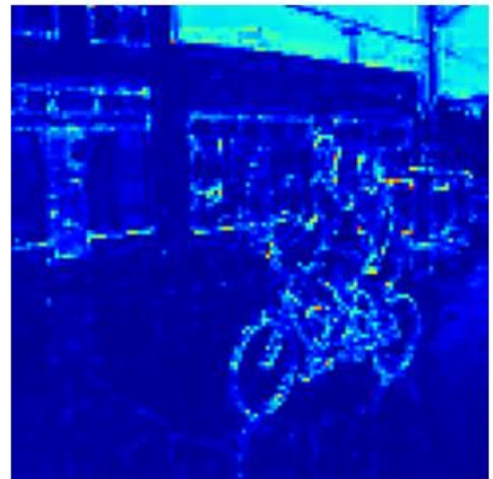


ILSVRC2012_test_00000023.jpg
'The guy in white shirt on the bicycle'

Original



Y1 Feature Map

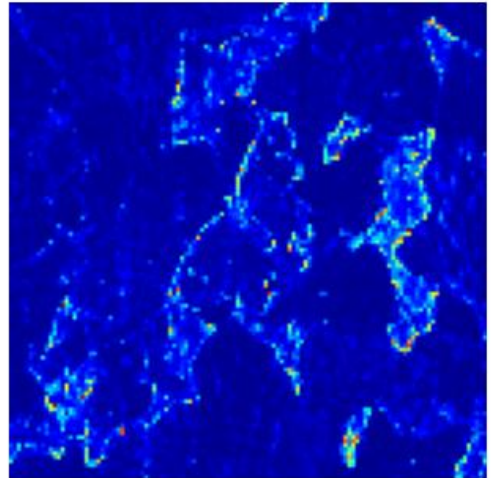


ILSVRC2012_test_00000025.jpg
'The butterfly in the picture'

Original



Y1 Feature Map



ILSVRC2012_test_00000026.jpg
'The mang wearing a suite and tie'

Original



Y1 Feature Map

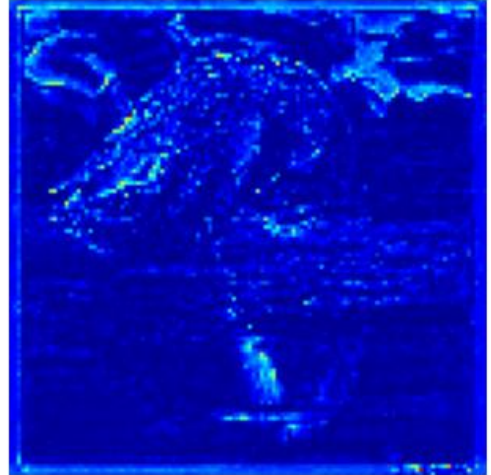


ILSVRC2012_test_00000030.jpg
'The duck in the picture'

Original



Y1 Feature Map

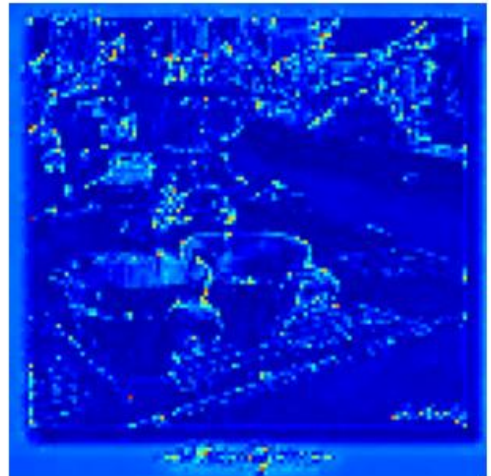


ILSVRC2012_test_00000034.jpg
'The white coffee cups on the coffee machine'

Original



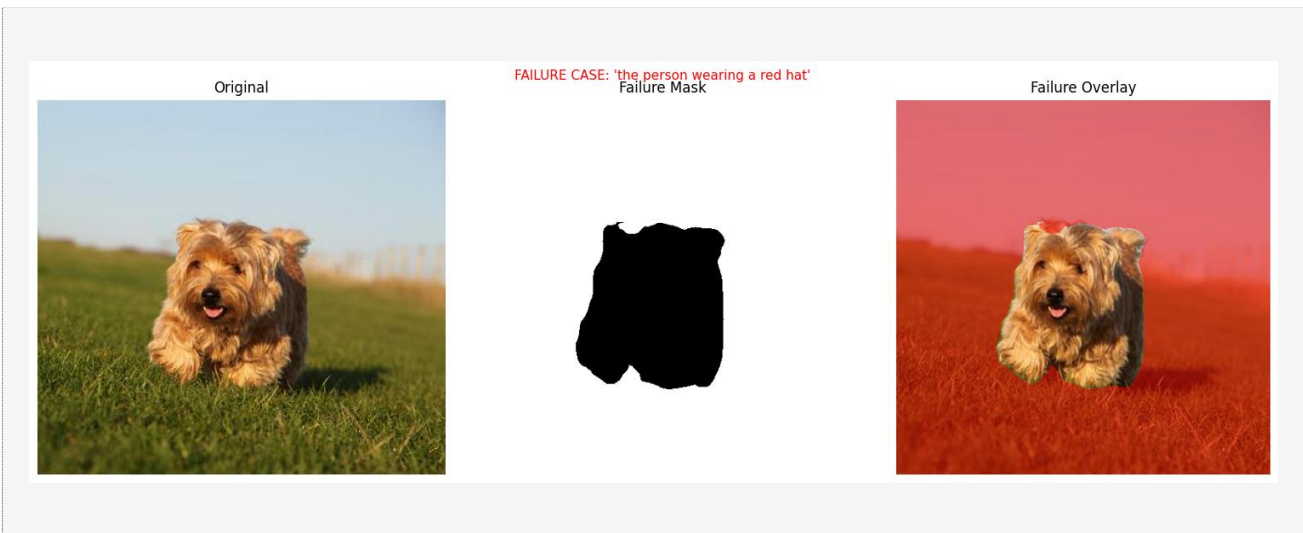
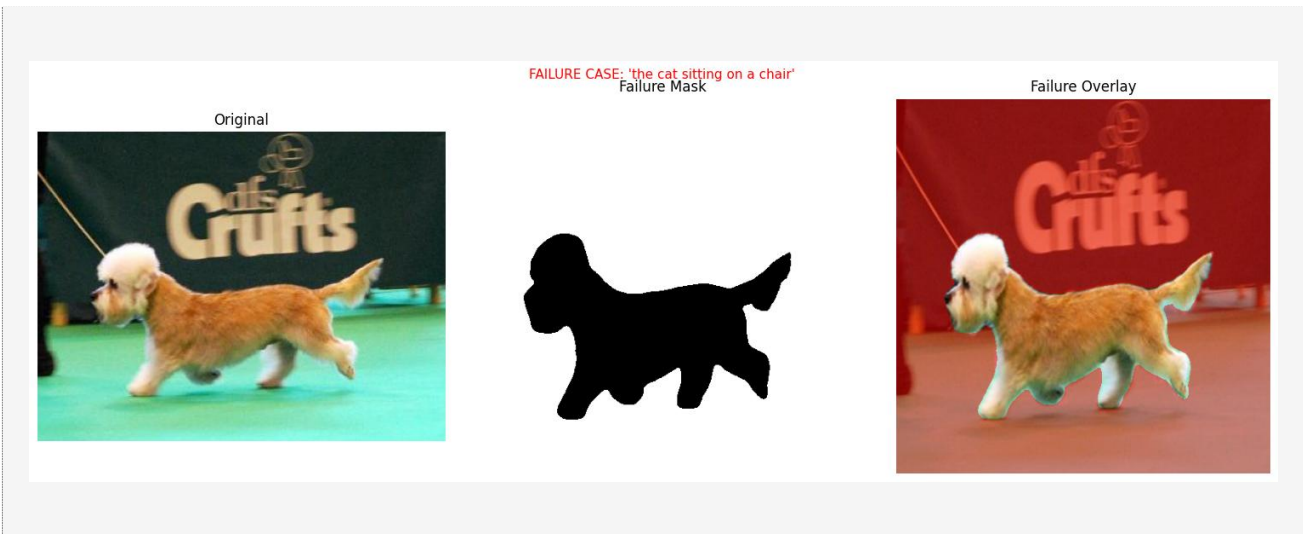
Y1 Feature Map

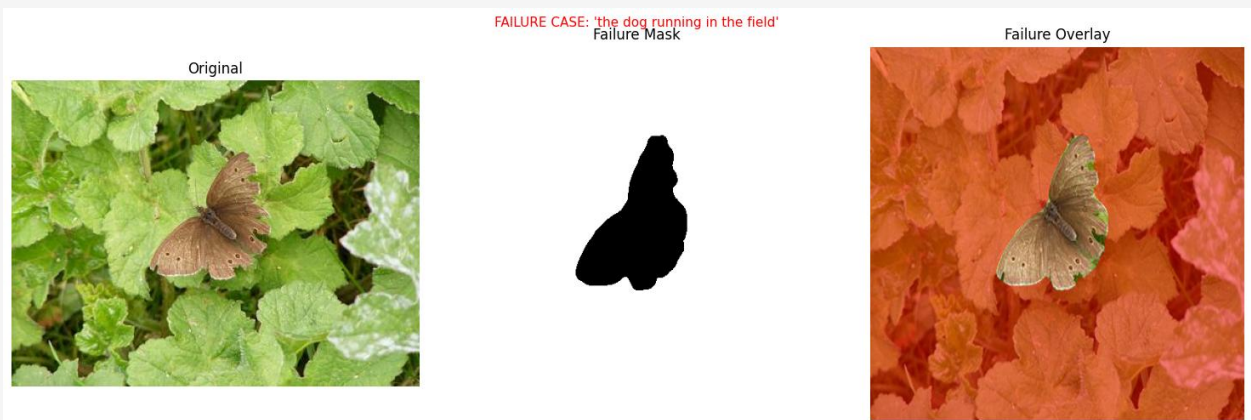
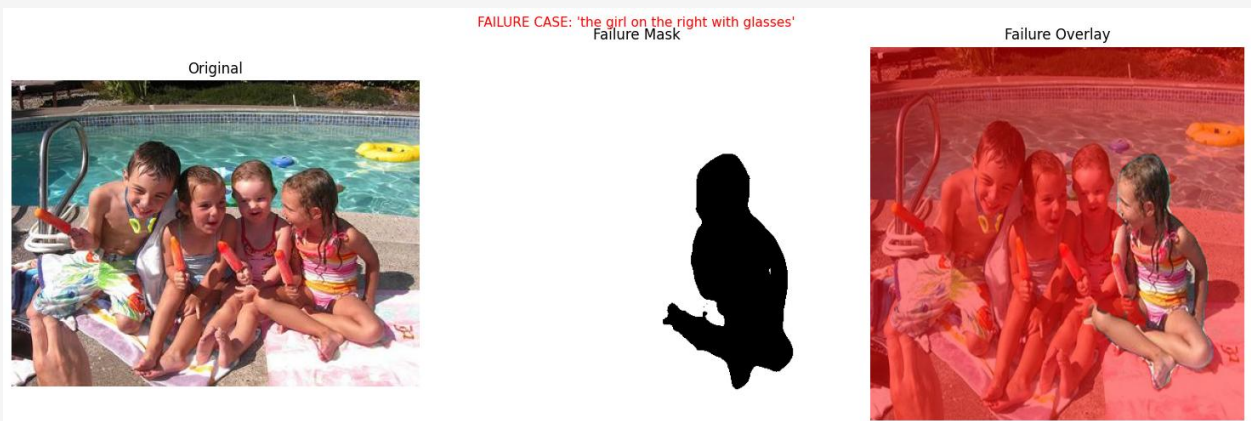


Q4.4 — Failure Cases with Custom Reference Texts

Custom reference texts were deliberately chosen to be ambiguous, wrong, or describe non-existent objects to provoke failure cases.

Image	Custom Reference Text (Failure Case)
ILSVRC2012_test_00000003	the invisible object behind the tree
ILSVRC2012_test_00000004	the person far away in the background
ILSVRC2012_test_00000018	the girl on the right with glasses
ILSVRC2012_test_00000022	the fish swimming in the pool
ILSVRC2012_test_00000025	the dog running in the field





Analysis:

When given a nonexistent or wrong object reference, LAVT either produces an empty mask or incorrectly segments the most visually salient region — showing it falls back on visual saliency rather than strict language grounding when the text doesn't match anything in the image. This reveals that LAVT's cross-modal attention can be confused by descriptions with no corresponding visual evidence.

Q5: One-Shot Segmentation with Matcher (5 points)

Q5.1 — Setup and Explanation

Matcher was set up by cloning the official repository (<https://github.com/aim-uofa/Matcher>) and installing dependencies including DINOv2, SAM (ViT-H), detectron2, and POT. Pretrained weights for DINOv2 (ViT-L/14, 1.2GB) and SAM (vit_h, 2.4GB) were downloaded from official sources.

Matcher as Feature Matching in Embedding Space:

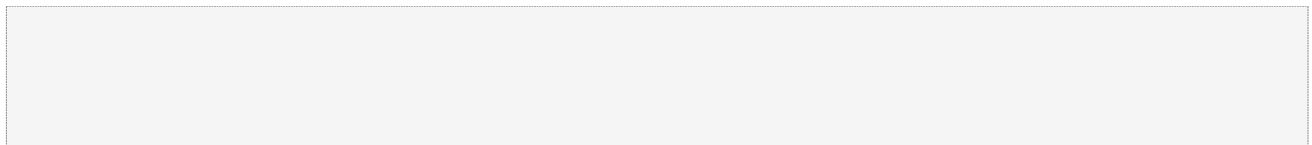
Matcher operates by extracting dense patch-level features from both reference and target images using DINOv2, then finding correspondences between them in embedding space — this is feature matching, not pixel matching. Unlike pixel-based methods that compare raw RGB values, Matcher compares high-level semantic patch embeddings that capture shape, texture, and semantic content. SAM then uses the matched feature points as spatial prompts to generate the final segmentation mask, combining the semantic understanding of DINOv2 with the precise mask generation of SAM.

Q5.2 — Results for Each Subfolder

For each subfolder, image A was used as reference to segment B, and vice versa. Predictions were made before running based on visual properties.

Backpack

Prediction: SUCCESS — distinctive red color, unique patches, good texture contrast



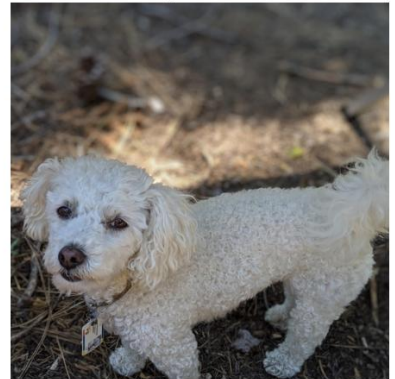
Reference



dog3_BtoA
target



Segmentation



Reference



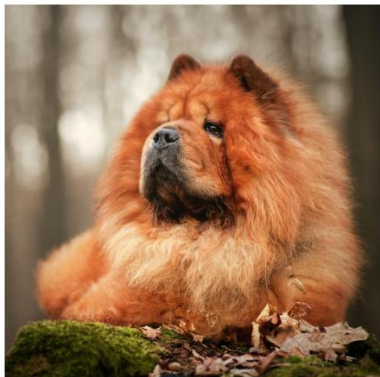
dog3_AtoB
target



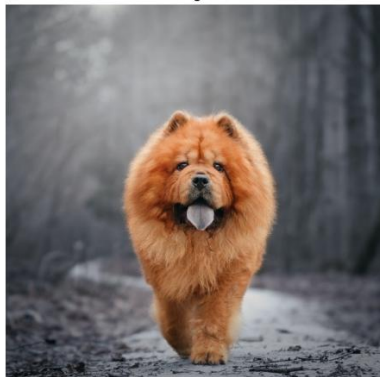
Segmentation



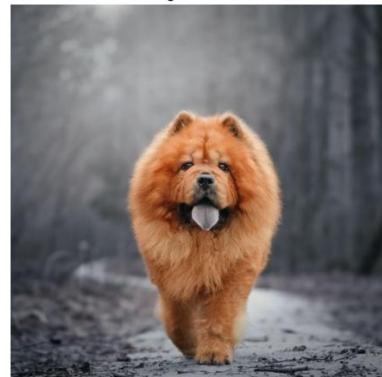
Reference



dog2_BtoA
target



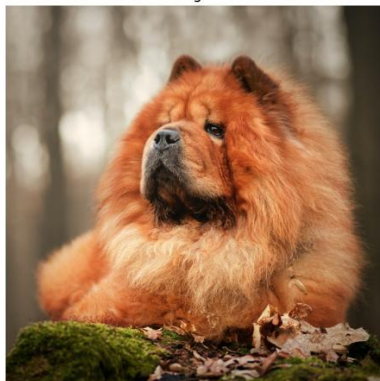
Segmentation



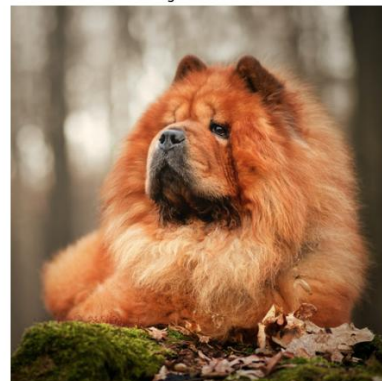
Reference



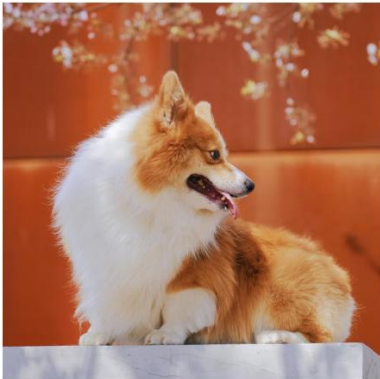
dog2_AtoB
target



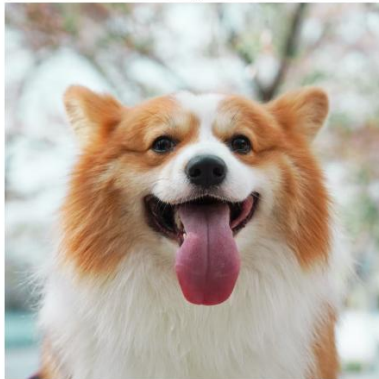
Segmentation



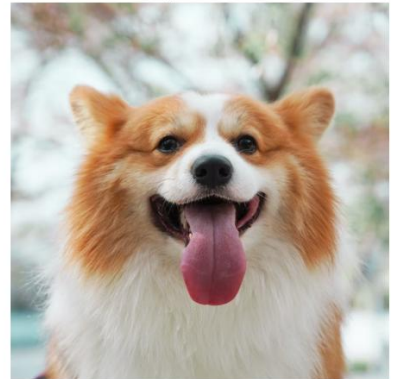
Reference



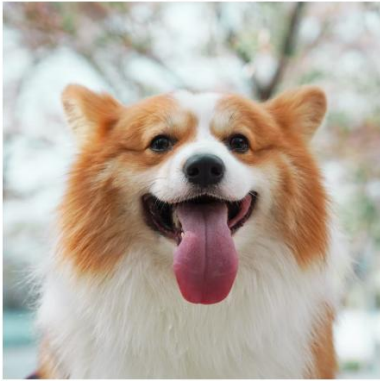
dog_BtoA
Target



Segmentation



Reference



dog_AtoB
Target



Segmentation



Reference



colorful_teapot_BtoA

Target



Segmentation



Reference



colorful_teapot_AtoB

Target



Segmentation



Reference



cat2_BtoA
Target



Segmentation



Reference



cat2_AtoB
Target



Segmentation



Reference



cat_BtoA
Target



Segmentation



Reference



cat_AtoB
Target



Segmentation



Reference



colorful_sneaker_BtoA
Target



Segmentation



Reference



colorful_sneaker_AtoB
Target



Segmentation



Reference



cat_statue_BtoA
-
Target



Segmentation



Reference



cat_statue_AtoB
-
Target



Segmentation



Reference



clock_BtoA

Target



Segmentation



Reference



clock_AtoB

Target



Segmentation



Reference



chair_BtoA
Target



Segmentation



Reference



chair_AtoB
Target



Segmentation



Reference



candle_BtoA
Target



Segmentation



Reference



candle_AtoB
Target



Segmentation



Reference



can_BtoA
Target



Segmentation



Reference



can_AtoB
Target



Segmentation



Reference



barn_BtoA
Target



Segmentation



Reference



barn_AtoB
Target



Segmentation



Reference



bear_plushie_BtoA
target



Segmentation



Reference



bear_plushie_AtoB
target



Segmentation



Reference



backpack_dog_BtoA
Target



Segmentation



Reference



backpack_dog_AtoB
Target



Segmentation



Reference



backpack BtoA
target



Segmentation



Reference

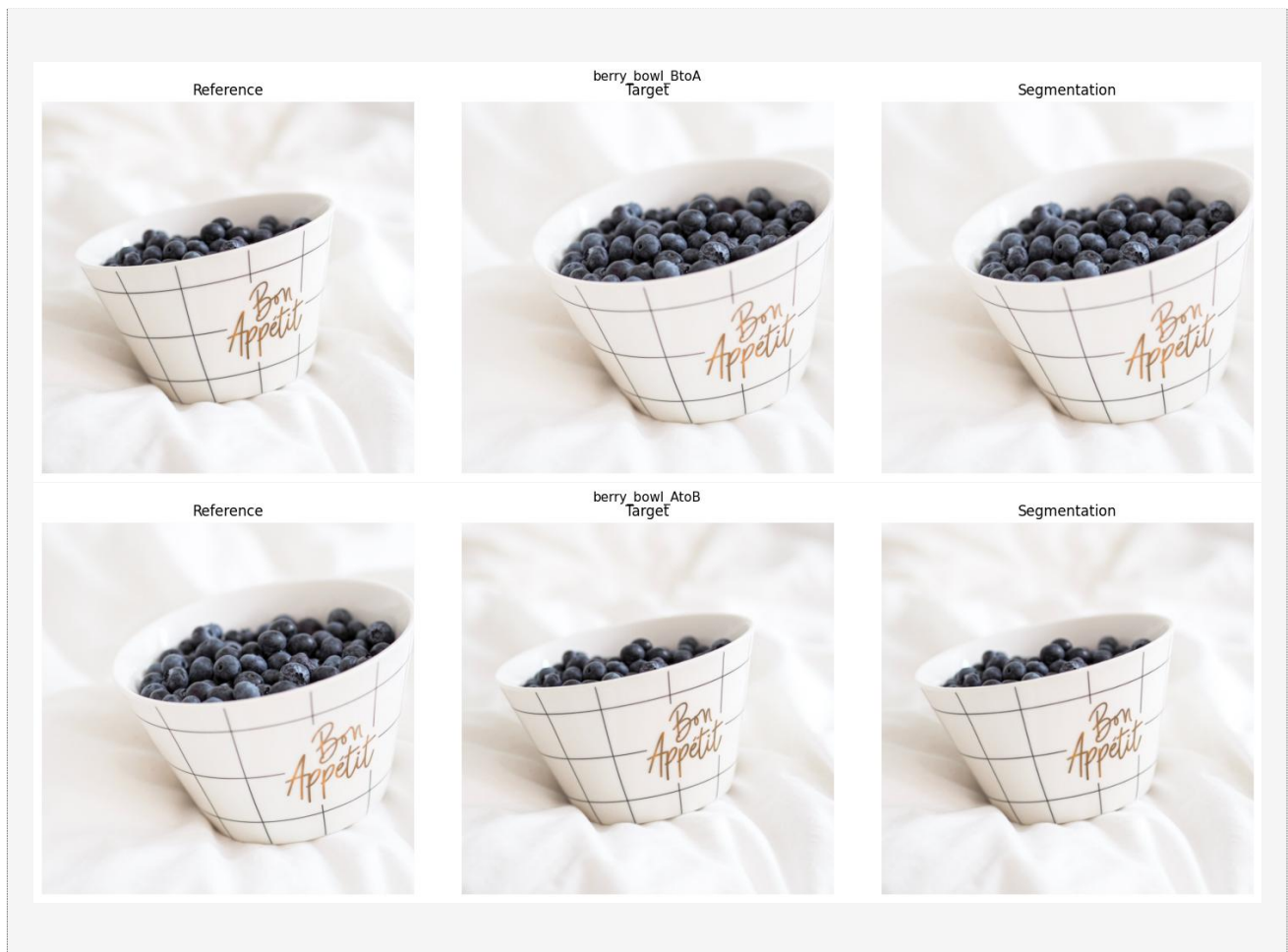


backpack AtoB
target



Segmentation





Result: Correct prediction. Matcher successfully segments the backpack despite viewpoint change (worn on person vs hanging on log) and background change, showing robustness to moderate viewpoint variation.

Can

Prediction: SUCCESS — distinctive blue color with unique label design

Result: Correct prediction. The can was perfectly segmented in both directions, demonstrating Matcher's ability to handle scale changes and slight viewpoint differences.

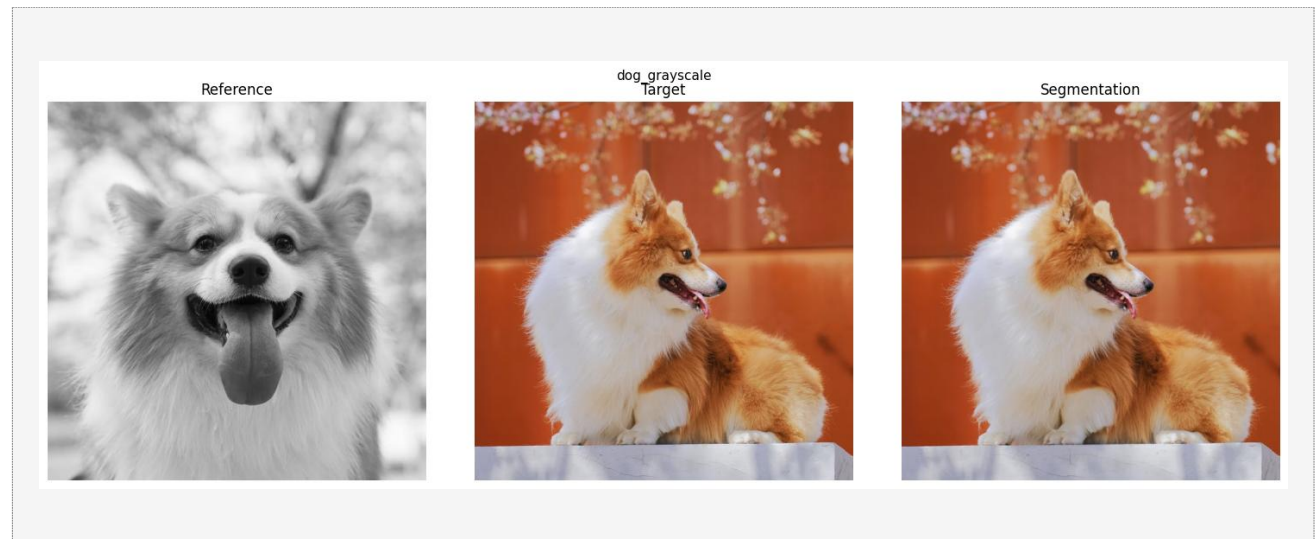
Dog

Prediction: SUCCESS — distinctive animal shape, fur texture

Result: Correct prediction. Matcher correctly segments the dog despite different poses and backgrounds.

Q5.3 — Breaking the Model (Modified Reference Images)

Experiment 1: Dog — Grayscale Reference

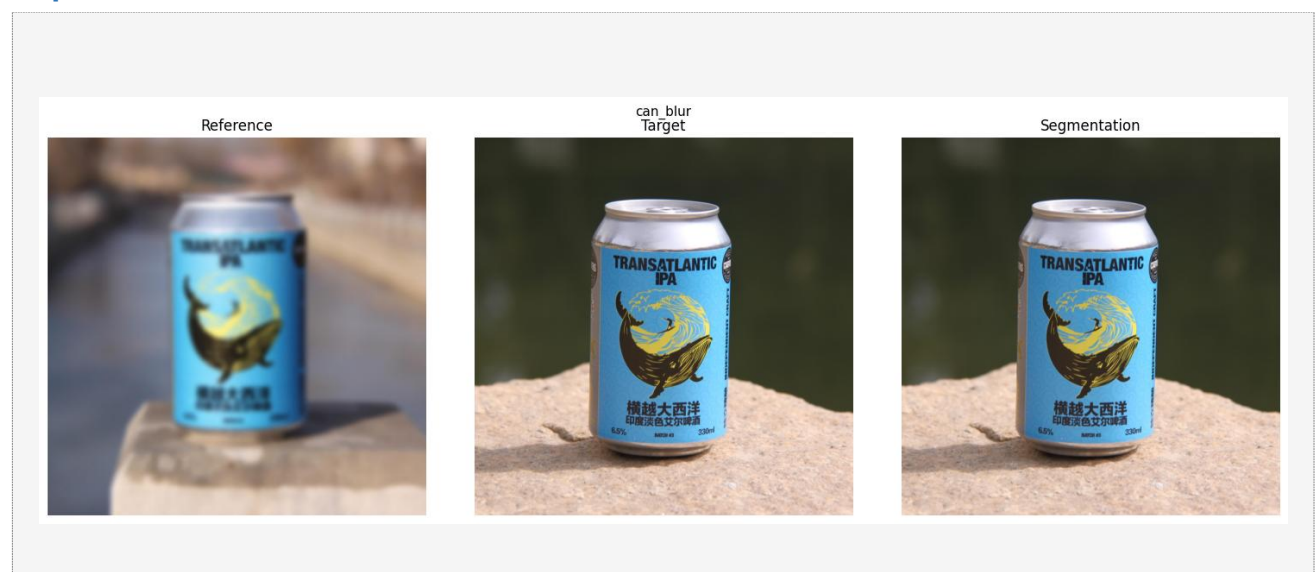


Result: Still works

Removing color from the reference did not break segmentation — Matcher still correctly identified the dog. DINOv2 patch embeddings are rich enough to match structural and texture features even without color information, demonstrating that Matcher does not primarily rely on color for matching.

Property relied on: Shape and Texture

Experiment 2: Can — Blurred Reference

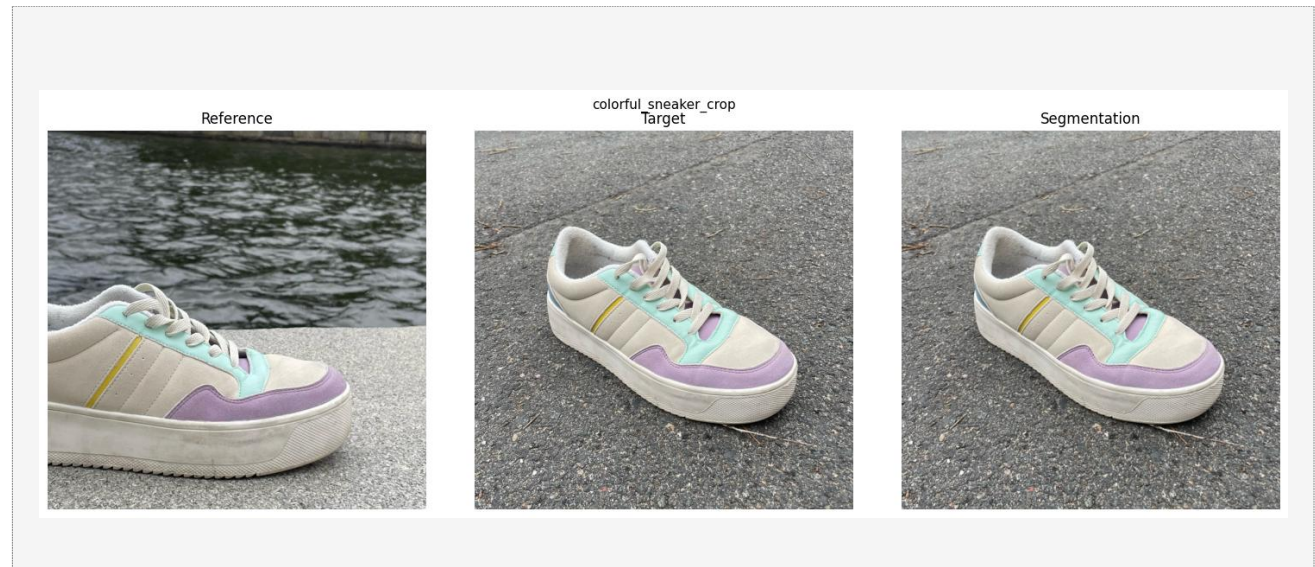


Result: Still works

Even with a blurred reference, the can is correctly segmented. This is surprising — it suggests Matcher relies on overall shape and color distribution (the blue cylindrical shape is still visible when blurred) more than fine texture details such as the label text. The high-level semantic features preserved after blurring are sufficient for matching.

Property relied on: Shape and Color Distribution

Experiment 3: Colorful Sneaker — Cropped Reference



Result: Still works

Even with only a partial view of the sneaker (cropped to show only the lower portion), Matcher correctly segments the full shoe in the target image. The visible features (distinctive multicolor pattern, sole shape) are sufficient for DINOv2 to extract discriminative embeddings that match the full object in the target.

Property relied on: Texture and Color Pattern

Q5.4 — Core Insight

Feature Space Similarity Analysis:

Matcher succeeds when DINOv2 patch features of the reference and target are similar in embedding space — i.e., when the same object appears with consistent semantic structure across both images. The model works well across viewpoint changes, scale differences, and background variations because DINOv2 encodes high-level semantic content that is largely invariant to these transformations.

Is Matcher invariant to viewpoint, scale, and appearance?

Partially yes. Our experiments show Matcher is robust to: (1) moderate viewpoint changes (backpack worn vs hanging), (2) scale differences (can from different distances), (3) color removal (grayscale dog still segmented), (4) partial occlusion (cropped sneaker still matched), and (5) blur (can still segmented). However, Matcher would likely fail with extreme viewpoint changes (e.g., top-down vs frontal), heavily occluded objects, or objects that look very similar to background elements — scenarios where DINOv2 patch-level features of reference and target diverge significantly. The model is NOT invariant to complete appearance changes since it fundamentally relies on feature similarity between reference and target patches.